

CPP102: PROGRAMMING LOGIC AND DESIGN

Week 4 Assessment Test Question Bank

Variables, Data Types, Type Conversion, and Operators

With Complete Instructor's Answer Key

Note to the Instructor

This document is an instructor's question bank. The instructor may select, revise, rearrange, or combine items to create different assessment test versions.

The question bank contains more items than a single test should include. Use the Suggested Test Combinations section at the end of this document as a starting point, or build a custom test by choosing items directly.

Topics covered: valid and invalid Python identifiers; variable declaration and assignment; data types (int, float, str, bool); type conversion using int(), float(), and str(); comments; arithmetic, comparison, logical, and assignment operators; operator precedence; input/processing/output; sequential Python programs; algorithm development; pseudocode; flowchart construction; predicting program output; and finding and correcting errors in Python code. All items use sequential programming logic only — conditional statements, loops, lists, functions, dictionaries, and file handling are not required.

SECTION A — Identifier and Variable Analysis

For items A1–A5, write VALID or INVALID for each identifier. For items A6–A10, write your complete answer in the space provided.

Item	Identifier	Difficulty	Points	VALID or INVALID
A1	quiz_score	Basic	1	
A2	3rd_attempt	Basic	1	
A3	student-name	Basic	1	
A4	total amount	Basic	1	
A5	return	Basic	1	

Problem A6 — Difficulty: Intermediate — Suggested Points: 2

Explain why "2nd_quiz" is an invalid identifier, then rewrite it as a valid identifier.

Answer:

Problem A7 — *Difficulty: Intermediate* — *Suggested Points: 2*

Explain why "electric-bill" is an invalid identifier, then rewrite it as a valid identifier.

Answer:

Problem A8 — *Difficulty: Intermediate* — *Suggested Points: 2*

Python identifiers are case-sensitive. Are "Distance" and "distance" the same variable? Explain your answer with a short example.

Answer:

Problem A9 — *Difficulty: Intermediate* — *Suggested Points: 2*

A program will record a student's name, the unit price of a school item, the quantity bought, and the total amount to be paid. Suggest one appropriate and valid Python identifier for each of these four values.

Answer:

Problem A10 — *Difficulty: Challenging* — *Suggested Points: 3*

A household electricity-monitoring program needs variables for: the previous meter reading, the current meter reading, the consumption in kilowatt-hours, and the total electric bill in pesos. Suggest a valid identifier for each of the four values, and identify which of the four is most likely to require the float data type. Explain why.

Answer:

SECTION B — Data Types and Type Conversion

Answer each item completely in the space provided.

Problem B1 — *Difficulty: Basic* — *Suggested Points: 1*

What data type does the value 25 belong to in Python?

Answer:

Problem B2 — *Difficulty: Basic* — *Suggested Points: 1*

What data type does the value 3.14 belong to in Python?

Answer:

Problem B3 — *Difficulty: Basic* — *Suggested Points: 1*

What data type does the value "Hello" belong to in Python?

Answer:

Problem B4 — *Difficulty: Basic* — *Suggested Points: 1*

What data type does the value True belong to in Python?

Answer:

Problem B5 — *Difficulty: Intermediate* — *Suggested Points: 2*

A program contains: age = input("Enter your age: "). If the user types 18, what is the data type of age, and why?

Answer:

Problem B6 — *Difficulty: Intermediate* — *Suggested Points: 2*

A program asks for the number of notebooks and their unit price, then multiplies them. Write the Python statements (with proper type conversion) so the two input values can be multiplied correctly.

Answer:

Problem B7 — *Difficulty: Intermediate* — *Suggested Points: 2*

A program has: name = "Ana" and age = 20, then executes print(name + age). Explain what happens, and correct the statement so it works.

Answer:

Problem B8 — *Difficulty: Challenging* — *Suggested Points: 3*

A payroll program has: hours = input("Enter hours worked: ") and rate = input("Enter hourly rate: "), then computes pay = hours * rate. Explain the error this produces, and rewrite the program so it computes correctly.

Answer:

Problem B9 — *Difficulty: Basic* — *Suggested Points: 1*

What data type does float(input(...)) always produce, as long as the user enters a valid number?

Answer:

SECTION C — Operators and Expression Evaluation

For items C1–C10, evaluate each expression and write the exact result. For items C11–C14, show your answer and briefly explain the order of operations where indicated.

Item	Expression	Difficulty	Points	Answer
C1	18 + 2 * 5	Basic	1	
C2	(18 + 2) * 5	Basic	1	
C3	29 // 6	Basic	1	
C4	29 % 6	Basic	1	
C5	4 ** 3	Basic	1	
C6	15 - 8 / 4	Intermediate	2	
C7	12 == 12 and 7 > 10	Intermediate	2	
C8	not (9 <= 3)	Intermediate	2	
C9	6 > 2 or 1 > 5	Intermediate	2	
C10	14 != 14	Basic	1	

Problem C11 — Difficulty: Challenging — Suggested Points: 3

```
x = 25
x -= 7
x *= 2
print(x)
```

Determine the final value of x and explain the order of operations.

Answer / Explanation:

Problem C12 — Difficulty: Intermediate — Suggested Points: 2

Using Python's floor division and modulo operators, find both the quotient and the remainder when 137 is divided by 12.

Answer / Explanation:

Problem C13 — *Difficulty: Challenging* — *Suggested Points: 3*

```
y = 10
y += 5
y //= 3
print(y)
```

Determine the final value of y and explain the order of operations.

Answer / Explanation:

Problem C14 — *Difficulty: Challenging* — *Suggested Points: 3*

```
2 + 3 ** 2 * 2
```

Answer / Explanation:

SECTION D — Predict the Program Output

Study each program carefully. Write the exact output that will appear in the terminal, including capitalization, spaces, decimal places, currency symbols, and True/False values.

Problem D1 — *Difficulty: Basic* — *Suggested Points: 2*

```
name = "Miguel"
section = "1CPE-B"
print(name)
print(section)
```

Expected Output:

Problem D2 — *Difficulty: Basic* — *Suggested Points: 2*

```
a = 9
b = 6
print(a - b)
print(a * b)
```

Expected Output:

Problem D3 — *Difficulty: Basic* — *Suggested Points: 2*

```
price = 48.75
qty = 2
total = price * qty
print(f"Total: ₱{total:.2f}")
```

Expected Output:

Problem D4 — *Difficulty: Intermediate* — *Suggested Points: 2*

```
value = "20"
result = int(value) + 15
print(result)
```

Expected Output:

Problem D5 — *Difficulty: Intermediate* — *Suggested Points: 2*

```
a = 53
b = 6
print(a // b)
print(a % b)
```

Expected Output:

Problem D6 — *Difficulty: Intermediate* — *Suggested Points: 2*

```
stock = 12
reorder_point = 15
needs_reorder = stock < reorder_point
print(needs_reorder)
```

Expected Output:

Problem D7 — *Difficulty: Intermediate* — *Suggested Points: 2*

```
has_id = True
has_uniform = True
can_enter = has_id and has_uniform
print(can_enter)
```

Expected Output:

Problem D8 — *Difficulty: Intermediate* — *Suggested Points: 2*

```
balance = 500
balance -= 125
print(balance)
```

Expected Output:

Problem D9 — *Difficulty: Challenging* — *Suggested Points: 3*

```
x = 3
y = 4
z = 2
result = x + y * z
print(result)
print(result - 1)
```

Expected Output:

Problem D10 — *Difficulty: Challenging* — *Suggested Points: 3*

```
rate = 8.25
hours = 6
pay = rate * hours
is_overtime = hours > 8
print(f"Pay: ₱{pay:.2f}")
print(is_overtime)
```

Expected Output:

SECTION E — Debugging and Code Correction

Each program below contains at least one error. Identify the error, then rewrite the program correctly in the space provided.

Problem E1 — *Difficulty: Intermediate* — *Suggested Points: 2*

`total-score = 82
print(total-score)`

Corrected Code and Explanation:

Problem E2 — *Difficulty: Basic* — *Suggested Points: 2*

`Print("Hello, Class!")`

Corrected Code and Explanation:

Problem E3 — *Difficulty: Basic* — *Suggested Points: 2*

`greeting = Welcome to CPP102
print(greeting)`

Corrected Code and Explanation:

Problem E4 — *Difficulty: Intermediate* — *Suggested Points: 2*

```
label = "Total: "  
amount = 150  
print(label + amount)
```

Corrected Code and Explanation:

Problem E5 — *Difficulty: Intermediate* — *Suggested Points: 2*

```
length = 8  
width = 5  
perimeter = length + width  
print(f"Perimeter: {perimeter}")
```

Corrected Code and Explanation:

Problem E6 — *Difficulty: Basic* — *Suggested Points: 2*

```
pricee = 45.00  
quantity = 3  
total = price * quantity  
print(total)
```

Corrected Code and Explanation:

Problem E7 — *Difficulty: Intermediate* — *Suggested Points: 2*

```
False = 10
print(False)
```

Corrected Code and Explanation:

Problem E8 — *Difficulty: Challenging* — *Suggested Points: 3*

```
hours = input("Enter hours worked: ")
rate = 55.0
pay = hours * rate
print(pay)
```

Corrected Code and Explanation:

SECTION F — Complete the Missing Code

Fill in each blank with the correct variable name, type-conversion function, operator, or statement.

Problem F1 — *Difficulty: Basic* — *Suggested Points: 2*

Total Price of Purchased Items

```
item_qty = _____(input("Enter quantity: "))           # Blank 1
item_price = _____(input("Enter unit price: "))        # Blank 2
total = item_qty _____ item_price                       # Blank 3
print(f"Total Price: ₱{total:.2f}")
```

Write the missing code above, then state your answers here:

Problem F2 — *Difficulty: Basic* — *Suggested Points: 2*

Student's Average Score

```
q1 = float(input("Enter first quiz score: "))
q2 = float(input("Enter second quiz score: "))
q3 = float(input("Enter third quiz score: "))
average = (q1 + q2 + q3) _____ 3          # Blank 1
print(_____ "Average: {average:.2f}")        # Blank 2
```

Write the missing code above, then state your answers here:

Problem F3 — *Difficulty: Intermediate* — *Suggested Points: 3*

Salary Based on Hours Worked

```
hours = _____(input("Enter hours worked: "))          # Blank 1
rate = _____(input("Enter hourly rate: "))             # Blank 2
gross_pay = hours _____ rate                            # Blank 3
print(f"Gross Pay: ₪{gross_pay:_____}")                  # Blank 4
```

Write the missing code above, then state your answers here:

Problem F4 — *Difficulty: Intermediate* — *Suggested Points: 2*

Travel Distance

```
speed = _____(input("Enter speed in km/h: "))          # Blank 1
time = float(input("Enter travel time in hours: "))
distance = speed _____ time                             # Blank 2
print(f"Distance: {distance:.2f} km")
```

Write the missing code above, then state your answers here:

Problem F5 — *Difficulty: Basic* — *Suggested Points: 2*

Celsius to Fahrenheit Conversion

```
celsius = float(input("Enter temperature in Celsius: "))
fahrenheit = (celsius _____ 9 / 5) _____ 32      # Blanks 1 and 2
print(f"Fahrenheit: {fahrenheit:.2f}")
```

Write the missing code above, then state your answers here:

Problem F6 — *Difficulty: Intermediate* — *Suggested Points: 2*

Remaining Student Allowance

```
allowance = float(input("Enter weekly allowance: "))
expenses = _____(input("Enter total expenses: "))      # Blank 1
remaining = allowance _____ expenses                    # Blank 2
print(f"Remaining: ₱{_____:.2f}")                          # Blank 3
```

Write the missing code above, then state your answers here:

Problem F7 — *Difficulty: Basic* — *Suggested Points: 2*

Area of a Rectangle

```
length = float(input("Enter length: "))
width = _____(input("Enter width: "))                  # Blank 1
area = length _____ width                               # Blank 2
print(f"Area: {area:.2f}")
```

Write the missing code above, then state your answers here:

Problem F8 — *Difficulty: Intermediate* — *Suggested Points: 2*

Change from Amount Paid

```
price = float(input("Enter the price: "))
amount_paid = _____(input("Enter amount paid: "))      # Blank 1
change = amount_paid _____ price                          # Blank 2
print(f"Change: ₱{_____:.2f}")                             # Blank 3
```

Write the missing code above, then state your answers here:

SECTION G — Full Problem-Solving Tasks

For each problem, prepare the following: an Input–Process–Output (IPO) table, an algorithm, pseudocode, a flowchart, a complete Python program, and the expected output using the given sample values. Use sequential programming logic only.

Problem G1 — *Difficulty: Basic* — *Suggested Points: 15*

School Supply Purchase

A student is buying supplies for the new school term at a bookstore. The program must ask for the number of reams of bond paper and the price per ream, the number of ballpens and the price per ballpen, and the number of folders and the price per folder. It must then compute and display the total purchase amount.

Given Values: Reams of bond paper: 2; Price per ream: ₱210.00; Ballpens: 5 pieces; Price per ballpen: ₱15.00; Folders: 10 pieces; Price per folder: ₱8.50

1. Input–Process–Output (IPO) Table

Input	Process	Output

2. Algorithm

Write your algorithm steps here:

3. Pseudocode

Write your pseudocode here:

4. Flowchart — Use standard flowchart symbols (terminator, parallelogram, rectangle, arrow). Follow Start → Input → Process → Output → End. Do not use decision diamonds.

Draw your flowchart in this space:

5. Python Program

Write your Python program here:

6. Expected Output (using the given values)

Expected Output:

Problem G2 — *Difficulty: Basic* — *Suggested Points: 12*

Average of Three Quiz Scores

A teacher wants a program that computes a student's average score in three quizzes. The program must ask for each of the three quiz scores and display the average.

Given Values: Quiz 1: 90; Quiz 2: 84; Quiz 3: 95

1. Input–Process–Output (IPO) Table

Input	Process	Output

2. Algorithm

Write your algorithm steps here:

3. Pseudocode

Write your pseudocode here:

4. Flowchart — Use standard flowchart symbols (terminator, parallelogram, rectangle, arrow). Follow Start → Input → Process → Output → End. Do not use decision diamonds.

Draw your flowchart in this space:

5. Python Program

Write your Python program here:

6. Expected Output (using the given values)

Expected Output:

Problem G3 — *Difficulty: Basic* — *Suggested Points: 10*

Celsius-to-Fahrenheit Conversion

A weather-monitoring program must convert a temperature reading from degrees Celsius to degrees Fahrenheit.

Given Values: Temperature: 24°C

1. Input–Process–Output (IPO) Table

Input	Process	Output

2. Algorithm

Write your algorithm steps here:

3. Pseudocode

Write your pseudocode here:

4. Flowchart — Use standard flowchart symbols (terminator, parallelogram, rectangle, arrow). Follow Start → Input → Process → Output → End. Do not use decision diamonds.

Draw your flowchart in this space:

5. Python Program

Write your Python program here:

6. Expected Output (using the given values)

Expected Output:

Problem G4 — *Difficulty: Intermediate* — *Suggested Points: 15*

Area and Perimeter of a Rectangular Garden

A homeowner is planning a rectangular garden and needs a program that computes both its area and perimeter, given the garden's length and width.

Given Values: Length: 9.5 meters; Width: 6.2 meters

1. Input–Process–Output (IPO) Table

Input	Process	Output

2. Algorithm

Write your algorithm steps here:

3. Pseudocode

Write your pseudocode here:

4. Flowchart — Use standard flowchart symbols (terminator, parallelogram, rectangle, arrow). Follow Start → Input → Process → Output → End. Do not use decision diamonds.

Draw your flowchart in this space:

5. Python Program

Write your Python program here:

6. Expected Output (using the given values)

Expected Output:

Problem G5 — *Difficulty: Intermediate* — *Suggested Points: 12*

Gross Salary Based on Hours Worked

A part-time employee is paid based on the number of hours worked in a week. The program must ask for the number of hours worked and the hourly rate, then compute the gross pay.

Given Values: Hours worked: 40; Hourly rate: ₱95.00

1. Input–Process–Output (IPO) Table

Input	Process	Output

2. Algorithm

Write your algorithm steps here:

3. Pseudocode

Write your pseudocode here:

4. Flowchart — Use standard flowchart symbols (terminator, parallelogram, rectangle, arrow). Follow Start → Input → Process → Output → End. Do not use decision diamonds.

Draw your flowchart in this space:

5. Python Program

Write your Python program here:

6. Expected Output (using the given values)

Expected Output:

Problem G6 — *Difficulty: Basic* — *Suggested Points: 10*

Change from an Amount Paid

A sari-sari store owner needs a program that computes the change to give a customer, given the price of an item and the amount the customer paid.

Given Values: Price: ₱458.75; Amount paid: ₱500.00

1. Input–Process–Output (IPO) Table

Input	Process	Output

2. Algorithm

Write your algorithm steps here:

3. Pseudocode

Write your pseudocode here:

4. Flowchart — Use standard flowchart symbols (terminator, parallelogram, rectangle, arrow). Follow Start → Input → Process → Output → End. Do not use decision diamonds.

Draw your flowchart in this space:

5. Python Program

Write your Python program here:

6. Expected Output (using the given values)

Expected Output:

Problem G7 — *Difficulty: Intermediate* — *Suggested Points: 15*

Remaining Student Allowance

A student wants to track a weekly allowance. The program must ask for the weekly allowance and the amounts spent on transportation fare, food, and mobile load, then compute the amount remaining.

Given Values: Weekly allowance: ₱350.00; Transportation fare: ₱60.00; Food: ₱120.00; Load: ₱50.00

1. Input–Process–Output (IPO) Table

Input	Process	Output

2. Algorithm

Write your algorithm steps here:

3. Pseudocode

Write your pseudocode here:

4. Flowchart — Use standard flowchart symbols (terminator, parallelogram, rectangle, arrow). Follow Start → Input → Process → Output → End. Do not use decision diamonds.

Draw your flowchart in this space:

5. Python Program

Write your Python program here:

6. Expected Output (using the given values)

Expected Output:

Problem G8 — *Difficulty: Basic* — *Suggested Points: 10*

Distance Traveled Using Speed and Time

A delivery rider wants to estimate the distance covered on a trip. The program must ask for the average speed and the travel time, then compute the distance traveled.

Given Values: Speed: 80 km/h; Time: 1.75 hours

1. Input–Process–Output (IPO) Table

Input	Process	Output

2. Algorithm

Write your algorithm steps here:

3. Pseudocode

Write your pseudocode here:

4. Flowchart — Use standard flowchart symbols (terminator, parallelogram, rectangle, arrow). Follow Start → Input → Process → Output → End. Do not use decision diamonds.

Draw your flowchart in this space:

5. Python Program

Write your Python program here:

6. Expected Output (using the given values)

Expected Output:

Problem G9 — *Difficulty: Challenging* — *Suggested Points: 18*

Simple Interest on a Savings Deposit

A bank customer wants to know how much simple interest a deposit will earn. The program must ask for the principal amount, the annual interest rate (in decimal form), and the time in years, then compute the interest earned and the total amount after interest.

Given Values: Principal: ₱15,000.00; Rate: 0.05 (5%); Time: 2 years

1. Input–Process–Output (IPO) Table

Input	Process	Output

2. Algorithm

Write your algorithm steps here:

3. Pseudocode

Write your pseudocode here:

4. Flowchart — Use standard flowchart symbols (terminator, parallelogram, rectangle, arrow). Follow Start → Input → Process → Output → End. Do not use decision diamonds.

Draw your flowchart in this space:

5. Python Program

Write your Python program here:

6. Expected Output (using the given values)

Expected Output:

Problem G10 — *Difficulty: Intermediate* — *Suggested Points: 12*

Monthly Electricity Consumption Cost

A household wants to estimate its monthly electric bill. The program must ask for the number of kilowatt-hours (kWh) consumed and the rate charged per kilowatt-hour, then compute the total cost.

Given Values: Consumption: 350 kWh; Rate: ₱11.50 per kWh

1. Input–Process–Output (IPO) Table

Input	Process	Output

2. Algorithm

Write your algorithm steps here:

3. Pseudocode

Write your pseudocode here:

4. Flowchart — Use standard flowchart symbols (terminator, parallelogram, rectangle, arrow). Follow Start → Input → Process → Output → End. Do not use decision diamonds.

Draw your flowchart in this space:

5. Python Program

Write your Python program here:

6. Expected Output (using the given values)

Expected Output:

Problem G11 — *Difficulty: Intermediate* — *Suggested Points: 12*

Total Cost of Computer Components

A student is assembling a budget computer and needs a program that adds up the prices of four components: the CPU, RAM, SSD, and monitor.

Given Values: CPU: ₱8,500.00; RAM: ₱2,200.00; SSD: ₱3,100.00; Monitor: ₱6,500.00

1. Input–Process–Output (IPO) Table

Input	Process	Output

2. Algorithm

Write your algorithm steps here:

3. Pseudocode

Write your pseudocode here:

4. Flowchart — Use standard flowchart symbols (terminator, parallelogram, rectangle, arrow). Follow Start → Input → Process → Output → End. Do not use decision diamonds.

Draw your flowchart in this space:

5. Python Program

Write your Python program here:

6. Expected Output (using the given values)

Expected Output:

Problem G12 — *Difficulty: Basic* — *Suggested Points: 10*

Converting Total Minutes into Hours and Minutes

A scheduling program needs to convert a duration given entirely in minutes into a more readable format of hours and remaining minutes.

Given Values: Total minutes: 257

1. Input–Process–Output (IPO) Table

Input	Process	Output

2. Algorithm

Write your algorithm steps here:

3. Pseudocode

Write your pseudocode here:

4. Flowchart — Use standard flowchart symbols (terminator, parallelogram, rectangle, arrow). Follow Start → Input → Process → Output → End. Do not use decision diamonds.

Draw your flowchart in this space:

5. Python Program

Write your Python program here:

6. Expected Output (using the given values)

Expected Output:

INSTRUCTOR'S COMPLETE ANSWER KEY

The answers below are organized using the same section and problem numbers as the question bank. This section may be omitted when printing the student copy.

Section A Answer Key — Identifier and Variable Analysis

Item	Identifier	VALID or INVALID	Reason, if invalid
A1	quiz_score	VALID	Starts with a letter, uses only letters and an underscore, and is not a reserved keyword.
A2	3rd_attempt	INVALID	Begins with a digit (3). Python identifiers cannot start with a number.
A3	student-name	INVALID	Contains a hyphen (-), which Python interprets as the subtraction operator.
A4	total amount	INVALID	Contains a space. Identifiers must be a single unbroken word (use total_amount instead).
A5	return	INVALID	"return" is a reserved Python keyword and cannot be used as a variable name.

- A6:** Invalid because it begins with the digit 2. Corrected: quiz_2 (or second_quiz).
Any acceptable rewrite that starts with a letter/underscore and contains no spaces or symbols is correct.
- A7:** Invalid because the hyphen is read as subtraction (electric - bill). Corrected: electric_bill.
Any acceptable rewrite using an underscore or camelCase instead of a hyphen is correct.
- A8:** No. Python treats "Distance" and "distance" as two completely different variables because identifiers are case-sensitive.
Example: Distance = 120 followed by print(distance) raises a NameError, because "distance" (lowercase) was never defined — only "Distance" was.
- A9:** Sample acceptable answers: student_name, unit_price, quantity, total_amount.
Any valid identifiers that clearly describe each value and follow naming rules should be accepted.
- A10:** Sample identifiers: previous_reading, current_reading, kwh_consumed, total_bill. The total_bill (and possibly kwh_consumed) most likely require float, since electric bills and metered consumption are usually expressed with decimal/centavo values.
Meter readings could reasonably be int or float depending on the meter; the total peso amount almost always needs float for centavos.

Section B Answer Key — Data Types and Type Conversion

- B1:**
int
- B2:**
float
- B3:**
str

B4:

bool

B5:

str (string).

The input() function always returns a string, no matter what the user types, unless it is explicitly converted using int() or float().

B6:

```

quantity = int(input("Enter number of notebooks: "))
unit_price = float(input("Enter unit price: "))
total = quantity * unit_price

```

Without converting the input() strings using int()/float(), the multiplication would raise a TypeError.

B7:

```

TypeError: can only concatenate str (not "int") to str. Corrected: print(name + str(age)) – or use an f-string: print(f"{name}{age}").

```

The + operator cannot combine a string with an integer directly; the number must first be converted to a string.

B8:

```

Since both hours and rate remain strings from input(), hours * rate attempts to repeat a string, raising: TypeError: can't multiply sequence by non-int of type 'str'.

```

*Correction: hours = float(input("Enter hours worked: ")) rate = float(input("Enter hourly rate: "))
pay = hours * rate*

B9:

float — regardless of whether the user types a whole number or a decimal number.

Section C Answer Key — Operators and Expression Evaluation

Item	Expression	Answer
C1	18 + 2 * 5	28
C2	(18 + 2) * 5	100
C3	29 // 6	4
C4	29 % 6	5
C5	4 ** 3	64
C6	15 - 8 / 4	13.0
C7	12 == 12 and 7 > 10	False
C8	not (9 <= 3)	True
C9	6 > 2 or 1 > 5	True
C10	14 != 14	False

C11: 36

*x -= 7 means x = x - 7 → 25 - 7 = 18. Then x *= 2 means x = x * 2 → 18 * 2 = 36. Assignment operators apply in the order they appear, top to bottom.*

C12: Quotient (137 // 12) = 11; Remainder (137 % 12) = 5.

C13: 5

$y += 5 \rightarrow y = 10 + 5 = 15$. Then $y //= 3 \rightarrow y = 15 // 3 = 5$.

C14: 20

Exponentiation is evaluated first ($3 ** 2 = 9$), then multiplication ($9 * 2 = 18$), then addition ($2 + 18 = 20$).

Section D Answer Key — Predict the Program Output

D1 Output:

Miguel
1CPE-B

D2 Output:

3
54

D3 Output:

Total: ₱97.50

D4 Output:

35

D5 Output:

8
5

D6 Output:

True

D7 Output:

True

D8 Output:

375

D9 Output:

11
10

D10 Output:

Pay: ₱49.50
False

Section E Answer Key — Debugging and Code Correction

E1 — Error: SyntaxError: cannot assign to expression here (the hyphen in total-score is read as subtraction).

Corrected Code:

total_score = 82
print(total_score)

Output: 82

E2 — Error: NameError: name 'Print' is not defined — Python's print() function must be written in all lowercase.

Corrected Code:

```
print("Hello, Class!")
```

Output: Hello, Class!

E3 — Error: SyntaxError: invalid syntax — the text has no quotation marks, so Python cannot read it as a string.

Corrected Code:

```
greeting = "Welcome to CPP102"
print(greeting)
```

Output: Welcome to CPP102

E4 — Error: TypeError: can only concatenate str (not "int") to str — a string cannot be joined to a number using +.

Corrected Code:

```
label = "Total: "
amount = 150
print(label + str(amount))
```

Output: Total: 150

E5 — Error: No crash, but the formula is wrong: perimeter should be 2 * (length + width), not length + width.

Corrected Code:

```
length = 8
width = 5
perimeter = 2 * (length + width)
print(f"Perimeter: {perimeter}")
```

Output: Perimeter: 26

E6 — Error: NameError: name 'price' is not defined — the variable was declared as pricee (typo) but referenced as price.

Corrected Code:

```
price = 45.00
quantity = 3
total = price * quantity
print(total)
```

Output: 135.0

E7 — Error: SyntaxError: cannot assign to False — True, False, and None are reserved keywords/constants in Python and can never be used as variable names.

Corrected Code:

```
exam_passed = 10
print(exam_passed)
```

Output: 10

E8 — Error: TypeError: can't multiply sequence by non-int of type 'float' — hours is still a string from input() and was never converted to a number.

Corrected Code:

```
hours = float(input("Enter hours worked: "))
rate = 55.0
pay = hours * rate
print(pay)
```

Output: (numeric pay value, e.g. 440.0 if hours = 8)

Section F Answer Key — Complete the Missing Code

F1 — Total Price of Purchased Items: Blank 1 = int Blank 2 = float Blank 3 = *

```
item_qty = int(input("Enter quantity: "))
item_price = float(input("Enter unit price: "))
total = item_qty * item_price
print(f"Total Price: ₱{total:.2f}")
```

Given: item_qty = 6, item_price = 18.75 → Expected Output: Total Price: ₱112.50

F2 — Student's Average Score: Blank 1 = / Blank 2 = f

```
q1 = float(input("Enter first quiz score: "))
q2 = float(input("Enter second quiz score: "))
q3 = float(input("Enter third quiz score: "))
average = (q1 + q2 + q3) / 3
print(f"Average: {average:.2f}")
```

Given: q1 = 88, q2 = 92, q3 = 79 → Expected Output: Average: 86.33

F3 — Salary Based on Hours Worked: Blank 1 = float Blank 2 = float Blank 3 = * Blank 4 = .2f

```
hours = float(input("Enter hours worked: "))
rate = float(input("Enter hourly rate: "))
gross_pay = hours * rate
print(f"Gross Pay: ₱{gross_pay:.2f}")
```

Given: hours = 22, rate = 65.00 → Expected Output: Gross Pay: ₱1430.00

F4 — Travel Distance: Blank 1 = float Blank 2 = *

```
speed = float(input("Enter speed in km/h: "))
time = float(input("Enter travel time in hours: "))
distance = speed * time
print(f"Distance: {distance:.2f} km")
```

Given: speed = 60.0, time = 2.5 → Expected Output: Distance: 150.00 km

F5 — Celsius to Fahrenheit Conversion: Blank 1 = * Blank 2 = +

```
celsius = float(input("Enter temperature in Celsius: "))
fahrenheit = (celsius * 9 / 5) + 32
print(f"Fahrenheit: {fahrenheit:.2f}")
```

Given: celsius = 30 → Expected Output: Fahrenheit: 86.00

F6 — Remaining Student Allowance: Blank 1 = float Blank 2 = - Blank 3 = remaining

```
allowance = float(input("Enter weekly allowance: "))
expenses = float(input("Enter total expenses: "))
remaining = allowance - expenses
print(f"Remaining: ₱{remaining:.2f}")
```

Given: allowance = 300.0, expenses = 175.50 → Expected Output: Remaining: ₱124.50

F7 — Area of a Rectangle: Blank 1 = float Blank 2 = *

```

length = float(input("Enter length: "))
width = float(input("Enter width: "))
area = length * width
print(f"Area: {area:.2f}")

```

Given: length = 12.5, width = 4.0 → Expected Output: Area: 50.00

F8 — Change from Amount Paid: Blank 1 = float Blank 2 = - Blank 3 = change

```

price = float(input("Enter the price: "))
amount_paid = float(input("Enter amount paid: "))
change = amount_paid - price
print(f"Change: ₱{change:.2f}")

```

Given: price = 235.00, amount_paid = 300.00 → Expected Output: Change: ₱65.00

Section G Answer Key — Full Problem-Solving Tasks

G1: School Supply Purchase

IPO Table

Input	Process	Output
Number and price of bond paper reams, ballpens, and folders	Compute the cost of paper, cost of pens, cost of folders, then the total purchase amount	Total purchase amount

Suggested Algorithm

1. Start.
2. Input the number and price of bond paper reams.
3. Input the number and price of ballpens.
4. Input the number and price of folders.
5. Compute the paper cost.
6. Compute the pen cost.
7. Compute the folder cost.
8. Compute the total purchase amount.
9. Display the total purchase amount.
10. End.

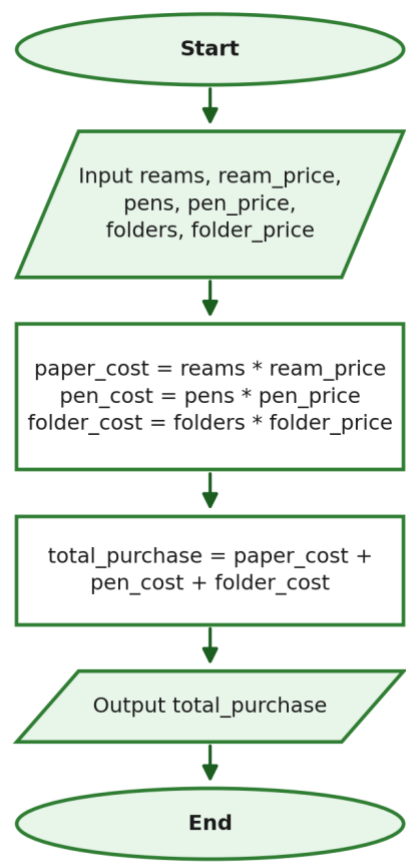
Suggested Pseudocode

```

START
INPUT reams, ream_price
INPUT pens, pen_price
INPUT folders, folder_price
paper_cost = reams * ream_price
pen_cost = pens * pen_price
folder_cost = folders * folder_price
total_purchase = paper_cost + pen_cost + folder_cost
OUTPUT total_purchase
END

```

Suggested Flowchart



Suggested Python Program

```
reams = int(input("Enter number of bond paper reams: "))
ream_price = float(input("Enter price per ream: "))
pens = int(input("Enter number of ballpens: "))
pen_price = float(input("Enter price per ballpen: "))
folders = int(input("Enter number of folders: "))
folder_price = float(input("Enter price per folder: "))

paper_cost = reams * ream_price
pen_cost = pens * pen_price
folder_cost = folders * folder_price
total_purchase = paper_cost + pen_cost + folder_cost

print(f"Total Purchase: ₱{total_purchase:.2f}")
```

Expected Output Using the Given Values

Total Purchase: ₱580.00

G2: Average of Three Quiz Scores

IPO Table

Input	Process	Output
Three quiz scores	Add the three scores and divide the sum by 3	Average score

Suggested Algorithm

- 1. Start.
- 2. Input the first quiz score.

- Input the second quiz score.
- Input the third quiz score.
- Compute the average of the three scores.
- Display the average.
- End.

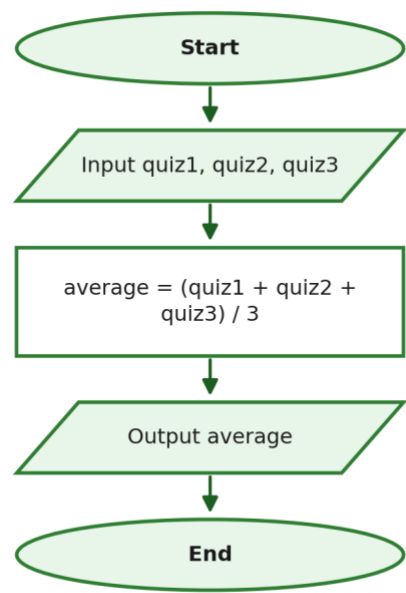
Suggested Pseudocode

```

START
INPUT quiz1
INPUT quiz2
INPUT quiz3
average = (quiz1 + quiz2 + quiz3) / 3
OUTPUT average
END

```

Suggested Flowchart



Suggested Python Program

```

quiz1 = float(input("Enter first quiz score: "))
quiz2 = float(input("Enter second quiz score: "))
quiz3 = float(input("Enter third quiz score: "))

average = (quiz1 + quiz2 + quiz3) / 3

print(f"Average: {average:.2f}")

```

Expected Output Using the Given Values

```

Average: 89.67

```

G3: Celsius-to-Fahrenheit Conversion

IPO Table

Input	Process	Output
Temperature in Celsius	Apply the conversion formula (celsius * 9/5) + 32	Temperature in Fahrenheit

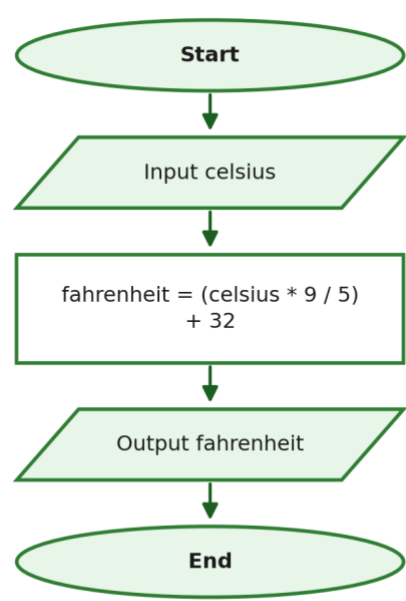
Suggested Algorithm

- 1. Start.
- 2. Input the temperature in Celsius.
- 3. Compute the equivalent temperature in Fahrenheit.
- 4. Display the temperature in Fahrenheit.
- 5. End.

Suggested Pseudocode

```
START
INPUT celsius
fahrenheit = (celsius * 9 / 5) + 32
OUTPUT fahrenheit
END
```

Suggested Flowchart



Suggested Python Program

```
celsius = float(input("Enter temperature in Celsius: "))

fahrenheit = (celsius * 9 / 5) + 32

print(f"Fahrenheit: {fahrenheit:.2f}")
```

Expected Output Using the Given Values

```
Fahrenheit: 75.20
```

G4: Area and Perimeter of a Rectangular Garden

IPO Table

Input	Process	Output
Length and width of the garden	Compute the area (length * width) and the perimeter (2 * (length + width))	Area and perimeter

Suggested Algorithm

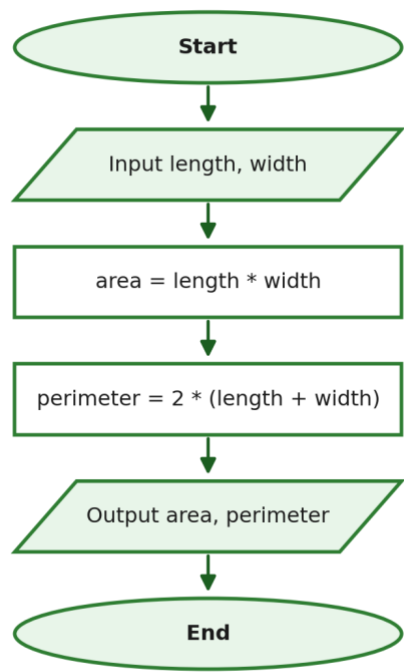
- 1. Start.

- 2. Input the length of the garden.
- 3. Input the width of the garden.
- 4. Compute the area.
- 5. Compute the perimeter.
- 6. Display the area and perimeter.
- 7. End.

Suggested Pseudocode

```
START
INPUT length
INPUT width
area = length * width
perimeter = 2 * (length + width)
OUTPUT area, perimeter
END
```

Suggested Flowchart



Suggested Python Program

```
length = float(input("Enter the length of the garden (m): "))
width = float(input("Enter the width of the garden (m): "))

area = length * width
perimeter = 2 * (length + width)

print(f"Area: {area:.2f} sq. m")
print(f"Perimeter: {perimeter:.2f} m")
```

Expected Output Using the Given Values

```
Area: 58.90 sq. m
Perimeter: 31.40 m
```

G5: Gross Salary Based on Hours Worked

IPO Table

Input	Process	Output
Hours worked and hourly rate	Multiply hours worked by the hourly rate	Gross pay

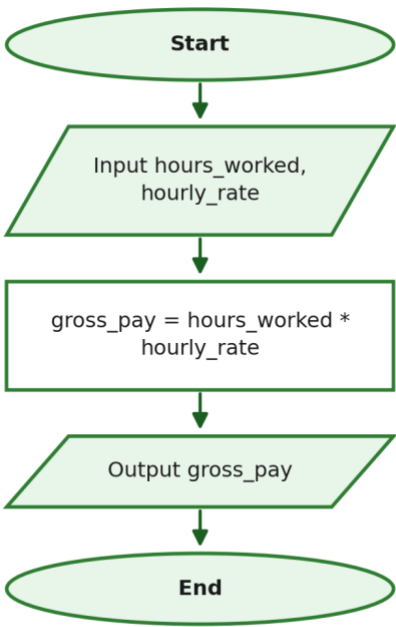
Suggested Algorithm

- 1. Start.
- 2. Input the number of hours worked.
- 3. Input the hourly rate.
- 4. Compute the gross pay.
- 5. Display the gross pay.
- 6. End.

Suggested Pseudocode

```
START
INPUT hours_worked
INPUT hourly_rate
gross_pay = hours_worked * hourly_rate
OUTPUT gross_pay
END
```

Suggested Flowchart



Suggested Python Program

```
hours_worked = float(input("Enter number of hours worked: "))
hourly_rate = float(input("Enter hourly rate: "))

gross_pay = hours_worked * hourly_rate

print(f"Gross Pay: ₱{gross_pay:.2f}")
```

Expected Output Using the Given Values

```
Gross Pay: ₱3800.00
```

G6: Change from an Amount Paid

IPO Table

Input	Process	Output
Price of the item and amount paid	Subtract the price from the amount paid	Change

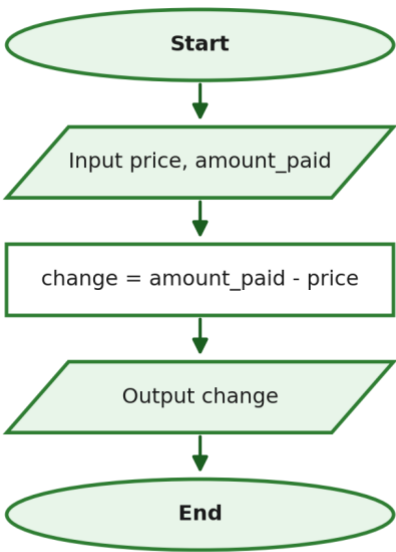
Suggested Algorithm

- 1. Start.
- 2. Input the price of the item.
- 3. Input the amount paid.
- 4. Compute the change.
- 5. Display the change.
- 6. End.

Suggested Pseudocode

```
START
INPUT price
INPUT amount_paid
change = amount_paid - price
OUTPUT change
END
```

Suggested Flowchart



Suggested Python Program

```
price = float(input("Enter the price of the item: "))
amount_paid = float(input("Enter the amount paid: "))

change = amount_paid - price

print(f"Change: ₱{change:.2f}")
```

Expected Output Using the Given Values

```
Change: ₱41.25
```

G7: Remaining Student Allowance

IPO Table

Input	Process	Output
Weekly allowance, fare, food, and load expenses	Subtract fare, food, and load from the allowance	Remaining allowance

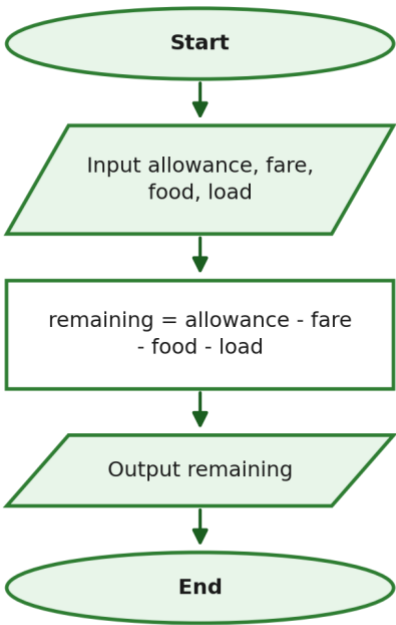
Suggested Algorithm

1. Start.
2. Input the weekly allowance.
3. Input the transportation fare expense.
4. Input the food expense.
5. Input the load expense.
6. Compute the remaining allowance.
7. Display the remaining allowance.
8. End.

Suggested Pseudocode

```
START
INPUT allowance
INPUT fare
INPUT food
INPUT load
remaining = allowance - fare - food - load
OUTPUT remaining
END
```

Suggested Flowchart



Suggested Python Program

```
allowance = float(input("Enter weekly allowance: "))
fare = float(input("Enter transportation expenses: "))
food = float(input("Enter food expenses: "))
load = float(input("Enter load/internet expenses: "))

remaining = allowance - fare - food - load

print(f"Remaining Allowance: ₱{remaining:.2f}")
```

Expected Output Using the Given Values

Remaining Allowance: ₱120.00

G8: Distance Traveled Using Speed and Time

IPO Table

Input	Process	Output
Average speed and travel time	Multiply speed by time	Distance traveled

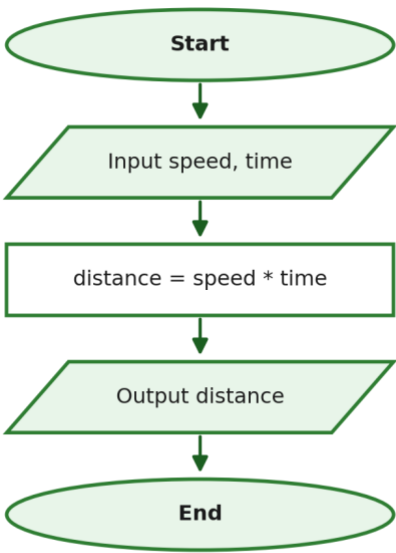
Suggested Algorithm

- 1. Start.
- 2. Input the average speed.
- 3. Input the travel time.
- 4. Compute the distance traveled.
- 5. Display the distance traveled.
- 6. End.

Suggested Pseudocode

```
START
INPUT speed
INPUT time
distance = speed * time
OUTPUT distance
END
```

Suggested Flowchart



Suggested Python Program

```
speed = float(input("Enter average speed (km/h): "))
time = float(input("Enter travel time (hours): "))

distance = speed * time

print(f"Distance: {distance:.2f} km")
```

Expected Output Using the Given Values

Distance: 140.00 km

G9: Simple Interest on a Savings Deposit

IPO Table

Input	Process	Output
Principal amount, interest rate, and time in years	Compute interest (principal * rate * time), then total amount (principal + interest)	Interest earned and total amount

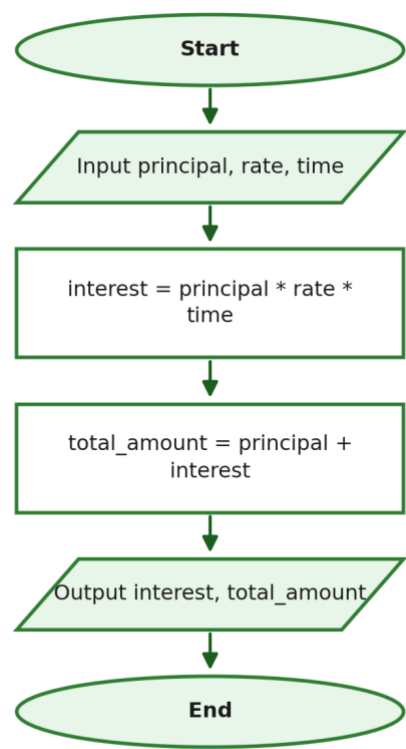
Suggested Algorithm

- 1. Start.
- 2. Input the principal amount.
- 3. Input the interest rate.
- 4. Input the time in years.
- 5. Compute the interest earned.
- 6. Compute the total amount.
- 7. Display the interest earned and the total amount.
- 8. End.

Suggested Pseudocode

```
START
INPUT principal
INPUT rate
INPUT time
interest = principal * rate * time
total_amount = principal + interest
OUTPUT interest, total_amount
END
```

Suggested Flowchart



Suggested Python Program

```
principal = float(input("Enter the principal amount: "))
rate = float(input("Enter the interest rate (decimal form): "))
time = float(input("Enter the time in years: "))

interest = principal * rate * time
total_amount = principal + interest

print(f"Interest: ₱{interest:.2f}")
print(f"Total Amount: ₱{total_amount:.2f}")
```

Expected Output Using the Given Values

```
Interest: ₱1500.00
Total Amount: ₱16500.00
```

G10: Monthly Electricity Consumption Cost

IPO Table

Input	Process	Output
Kilowatt-hours consumed and rate per kilowatt-hour	Multiply consumption by the rate	Total electricity cost

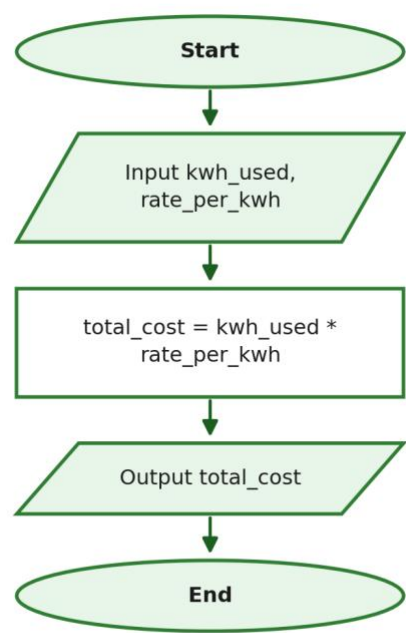
Suggested Algorithm

1. Start.
2. Input the kilowatt-hours consumed.
3. Input the rate per kilowatt-hour.
4. Compute the total electricity cost.
5. Display the total electricity cost.
6. End.

Suggested Pseudocode

```
START
INPUT kwh_used
INPUT rate_per_kwh
total_cost = kwh_used * rate_per_kwh
OUTPUT total_cost
END
```

Suggested Flowchart



Suggested Python Program

```
kwh_used = float(input("Enter kilowatt-hours used: "))
rate_per_kwh = float(input("Enter rate per kilowatt-hour: "))

total_cost = kwh_used * rate_per_kwh

print(f"Electricity Cost: ₱{total_cost:.2f}")
```

Expected Output Using the Given Values

```
Electricity Cost: ₱4025.00
```

G11: Total Cost of Computer Components

IPO Table

Input	Process	Output
Prices of the CPU, RAM, SSD, and monitor	Add the four component prices together	Total cost

Suggested Algorithm

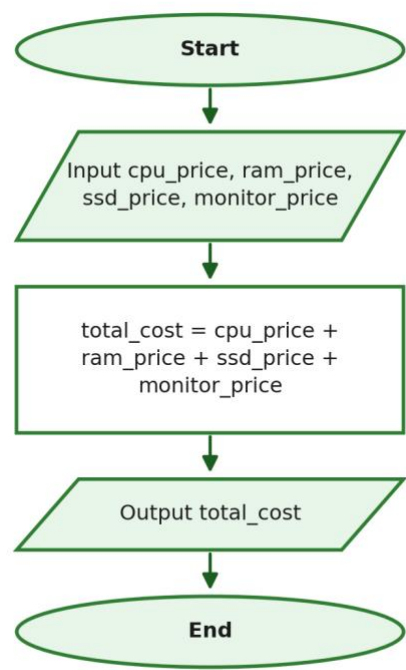
- 1. Start.
- 2. Input the CPU price.
- 3. Input the RAM price.
- 4. Input the SSD price.
- 5. Input the monitor price.
- 6. Compute the total cost.

- 7. Display the total cost.
- 8. End.

Suggested Pseudocode

```
START
INPUT cpu_price
INPUT ram_price
INPUT ssd_price
INPUT monitor_price
total_cost = cpu_price + ram_price + ssd_price + monitor_price
OUTPUT total_cost
END
```

Suggested Flowchart



Suggested Python Program

```
cpu_price = float(input("Enter CPU price: "))
ram_price = float(input("Enter RAM price: "))
ssd_price = float(input("Enter SSD price: "))
monitor_price = float(input("Enter monitor price: "))

total_cost = cpu_price + ram_price + ssd_price + monitor_price

print(f"Total Cost: ₱{total_cost:.2f}")
```

Expected Output Using the Given Values

```
Total Cost: ₱20300.00
```

G12: Converting Total Minutes into Hours and Minutes

IPO Table

Input	Process	Output
Total number of minutes	Use floor division to get whole hours; use modulo to get remaining minutes	Hours and remaining minutes

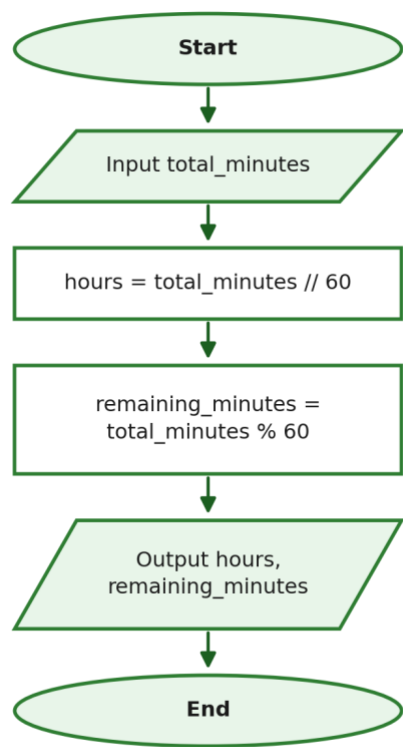
Suggested Algorithm

- 1. Start.
- 2. Input the total number of minutes.
- 3. Compute the number of whole hours.
- 4. Compute the remaining minutes.
- 5. Display the hours and remaining minutes.
- 6. End.

Suggested Pseudocode

```
START
INPUT total_minutes
hours = total_minutes // 60
remaining_minutes = total_minutes % 60
OUTPUT hours, remaining_minutes
END
```

Suggested Flowchart



Suggested Python Program

```
total_minutes = int(input("Enter total minutes: "))

hours = total_minutes // 60
remaining_minutes = total_minutes % 60

print(f"Hours: {hours}")
print(f"Minutes: {remaining_minutes}")
```

Expected Output Using the Given Values

```
Hours: 4
Minutes: 17
```

Suggested Test Combinations

The item numbers below refer to the problems in the question bank above. Each version draws from different items so instructors can prepare parallel test forms for different sections.

Suggested Version A — Appropriate for Beginners (Approx. 40–50 points)

Mixture of short problems and one full programming problem.

Part	Selected Item Numbers	Points
Identifier Analysis	A1–A5	5
Data Types / Conversion	B1–B4, B9	5
Expression Evaluation	C1–C5, C10	6
Program Output	D1, D2, D3	6
Code Correction	E2, E3	4
Complete the Code	F1, F5	4
Full Problem Solving	G3	10
Full Problem Solving	G8	10
Total		50

Suggested Version B — Moderate Difficulty (Approx. 40–50 points)

Different items from Version A.

Part	Selected Item Numbers	Points
Identifier Analysis	A6, A7, A9	6
Data Types / Conversion	B5, B6, B7	6
Expression Evaluation	C6, C7, C8, C9	8
Program Output	D4, D5, D6, D7	8
Code Correction	E1, E5, E6	6
Complete the Code	F3, F6	5
Full Problem Solving	G6	10
Total		49

Suggested Version C — Moderate to Challenging (Approx. 40–50 points)

Different items from Versions A and B.

Part	Selected Item Numbers	Points
Identifier Analysis	A8, A10	5
Data Types / Conversion	B8	3
Expression Evaluation	C11, C12, C13	8
Program Output	D8, D9, D10	8
Code Correction	E4, E7	4
Complete the Code	F4, F8	4
Full Problem Solving	G9	18
Total		50